

Problem

In a parallel RLC circuit, the bandwidth B is directly proportional to R .

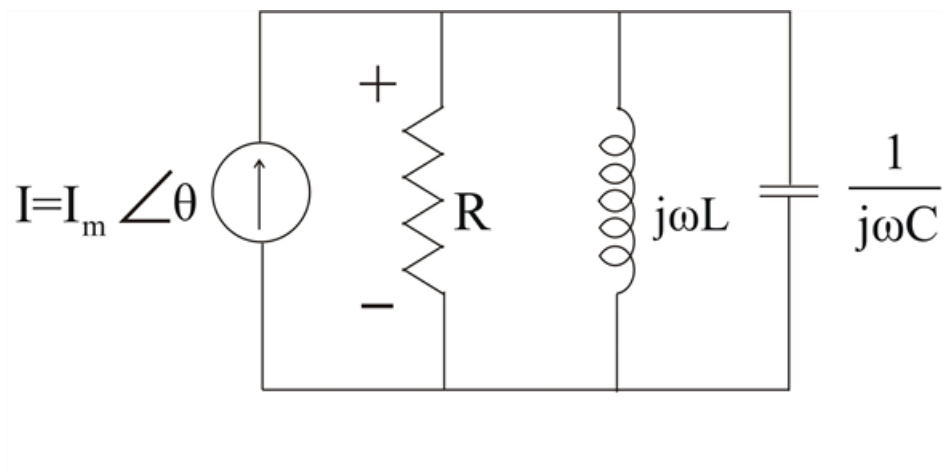
- (a) True
(b) False

Step-by-step solution

Step 1 of 3

The parallel resonant circuit is

Step 2 of 3



Step 3 of 3

The admittance is $Y = H(\omega) = \frac{I}{V}$ —(1)

$$\text{or } Y = \frac{1}{R} + j\omega C + \frac{1}{j\omega L}$$

$$Y = \frac{1}{R} + j\left(\omega L - \frac{1}{\omega C}\right) \quad \text{—(2)}$$

At resonance, $\omega = \omega_o$, and imaginary part of $Y = 0$.

$$\omega L - \frac{1}{\omega C} = 0 \Rightarrow \omega = \omega_o$$

$$\omega_o = \frac{1}{\sqrt{LC}} \quad \text{—(3)}$$

$$\text{Band width} \rightarrow B = \omega_2 - \omega_1 = \frac{1}{RC}$$

$$\text{or } B = \frac{1}{RC} \quad \text{—(4)}$$

$$B \propto \frac{1}{R}$$

Ans false, , because, B is not directly proportional to R .